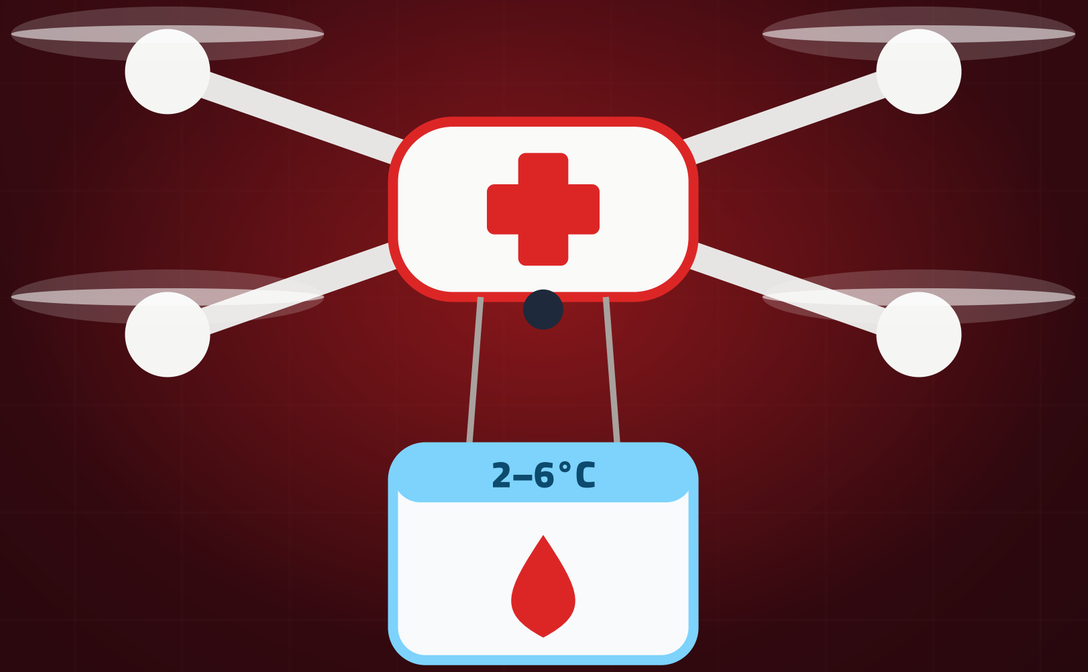
DAMI

Blood that arrives… before it's too late

An AI platform and medical drones for managing and delivering blood in emergencies

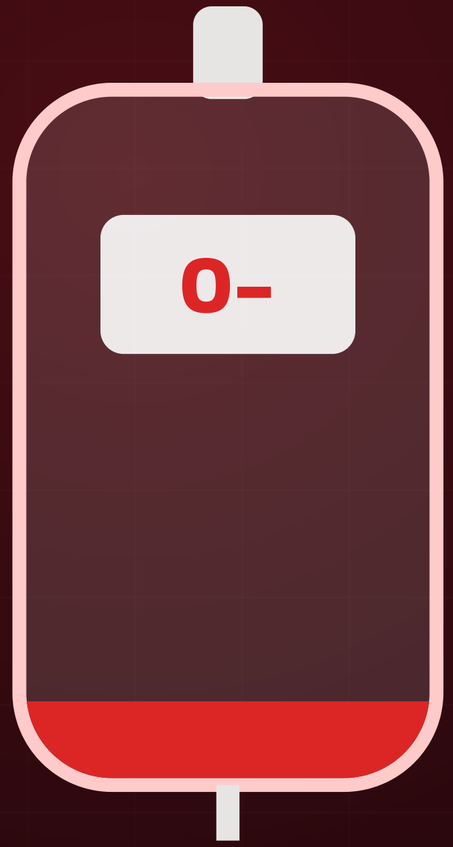


Team DAMI · Software Engineering & Artificial Intelligence

30–40%

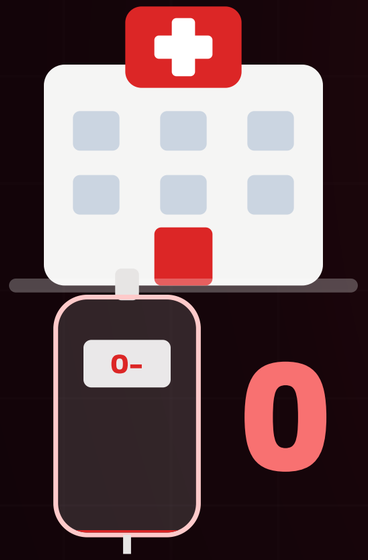
of trauma deaths are caused by bleeding

and every minute of delay raises the risk

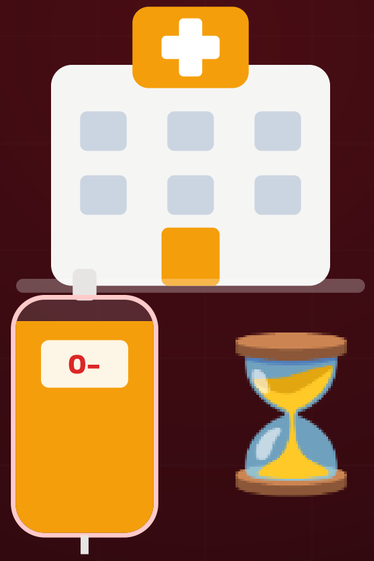


Source: Kauvar et al., Journal of Trauma, 2006

Shortage here... waste there



One hospital needs O– and has none



while units expire at another

5–7 days
platelet shelf life

Traffic
delays emergency transport

Generic drives
miss the missing type

Why does blood run short?



Seasonal donation

drops in summer, Ramadan and exams



Short shelf life

platelets 5–7 days, red cells 35–42



Rare types

O– always needed, rarely available



Uneven distribution

stock in the center, outskirts wait



Sudden emergencies

one accident uses days of stock



Manual coordination

phone calls, no live view

The solution: DAMI



Forecast & balance

before shortage or waste



Call the donor

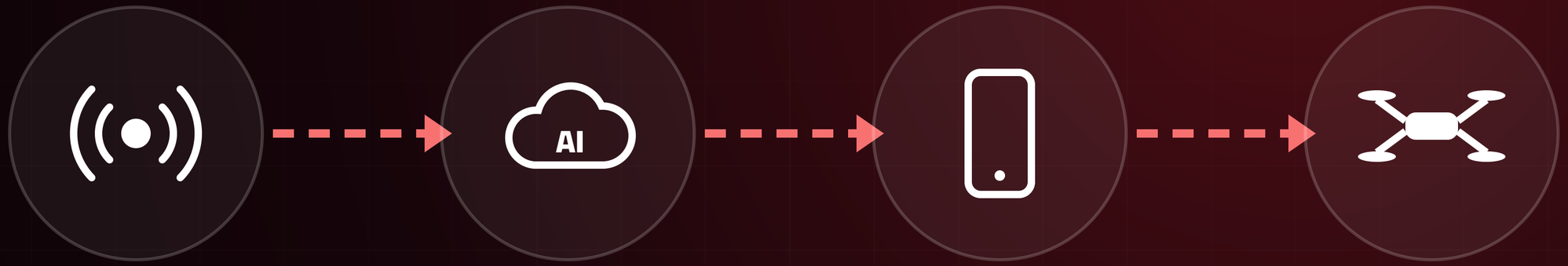
the right one, nearby



Blood drone

in minutes

How the system works



Blood-bank sensors

temperature + live stock

DAMI AI cloud

forecast + best source

Hospital app

one-tap emergency

Cooled drone

GPS + live temperature

The system in depth: 8 smart modules

🏷️

Smart inventory

RFID per unit with expiry

🧠

Demand forecasting

per type & hospital, 14 days

⚖️

Smart balancing

move units before they expire

🔍

Matching engine

compatible type, nearest source

🚁

Fleet management

safe route, licensed pilot approval

🌡️

Cold chain

temperature every second, alarms

🫂

Donor engine

targeted call-up for missing type

🏛️

National dashboard

live view for the Ministry

Emergency request: from tap to arrival

0 s
Hospital requests 6 units O−

5 s
AI checks stock & compatibility

10 s
Best source: drone or car

1 min
Pilot approval, box packed

≈ 9 min
Flight with live temperature & GPS

✓
Delivered, donors called to restock

Times from the simulation for a typical distance in Amman

The blood ambulance drone

2–6 °C box

temperature every second

4–6 units

per flight

Minutes

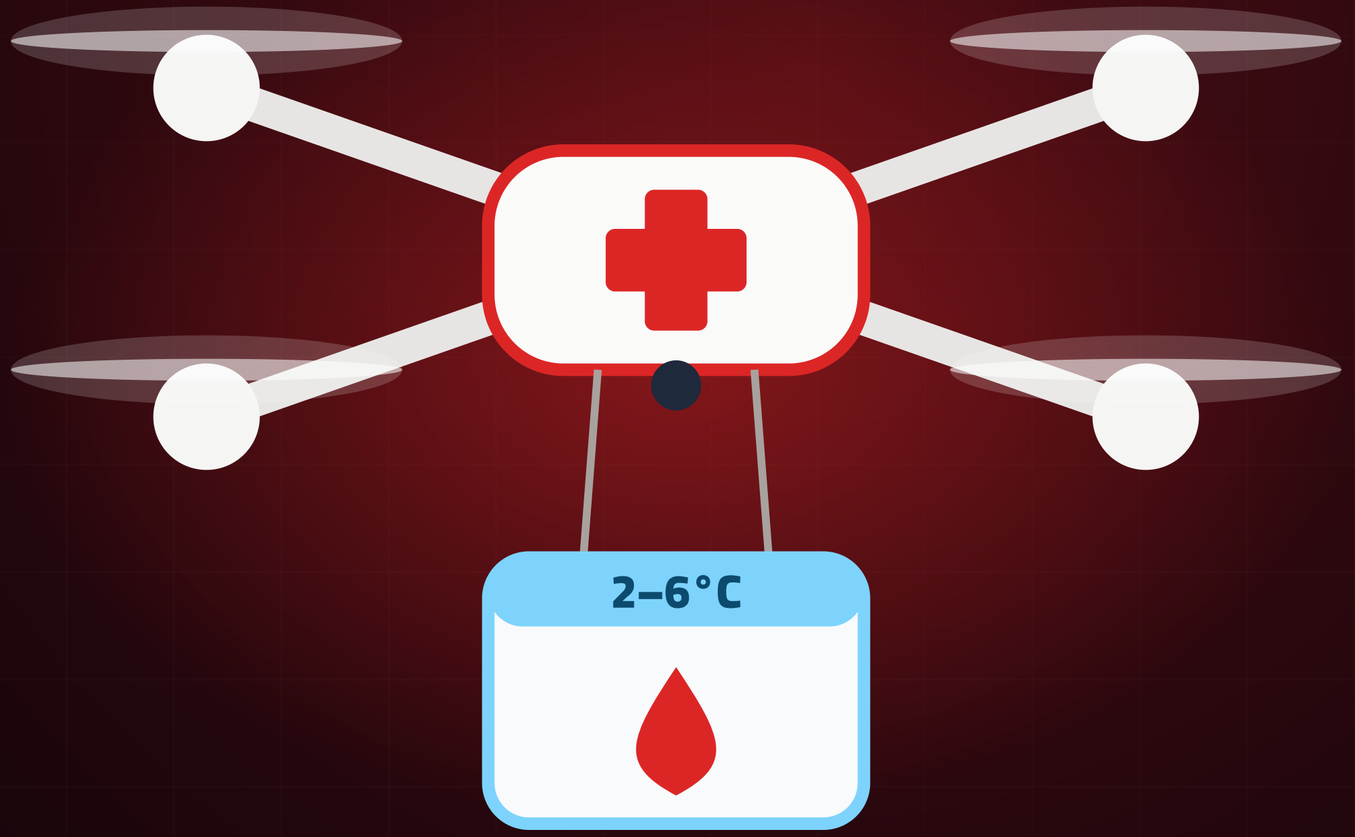
not half an hour

Live tracking

GPS, safe route

Safety

parachute, auto-return



Sensors & the cold chain

Blood-bank fridge

temperature & live stock sensor

Every blood unit

ID tag and expiry date

Drone box

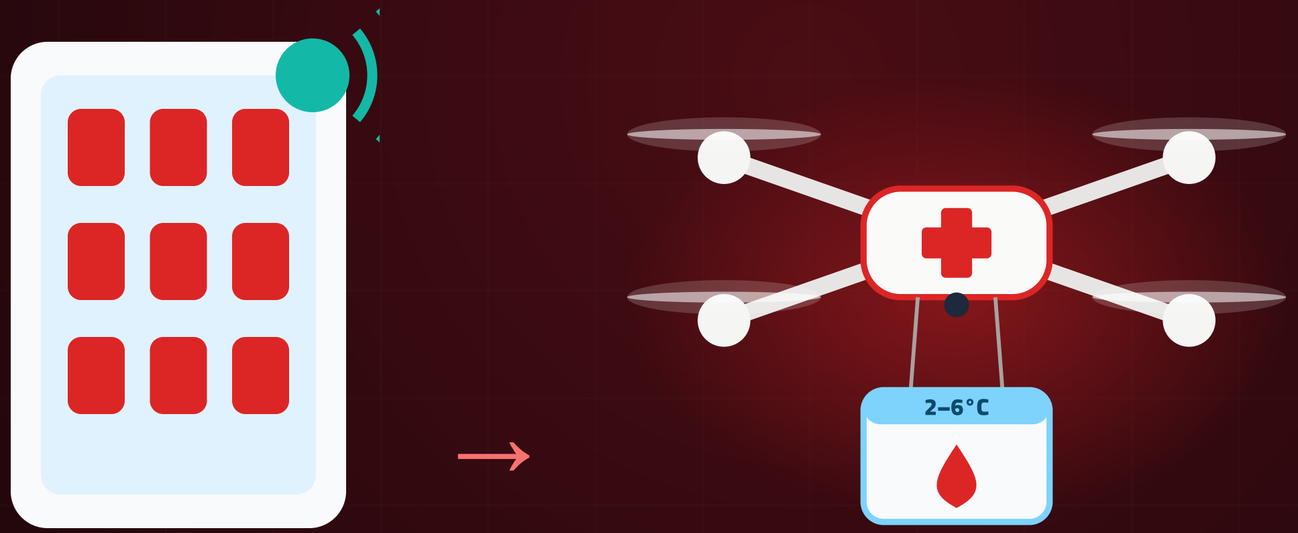
2–6°C every second + GPS

Instant alarm

backup cooling, hospital notified

Full log

temperature report with every shipment

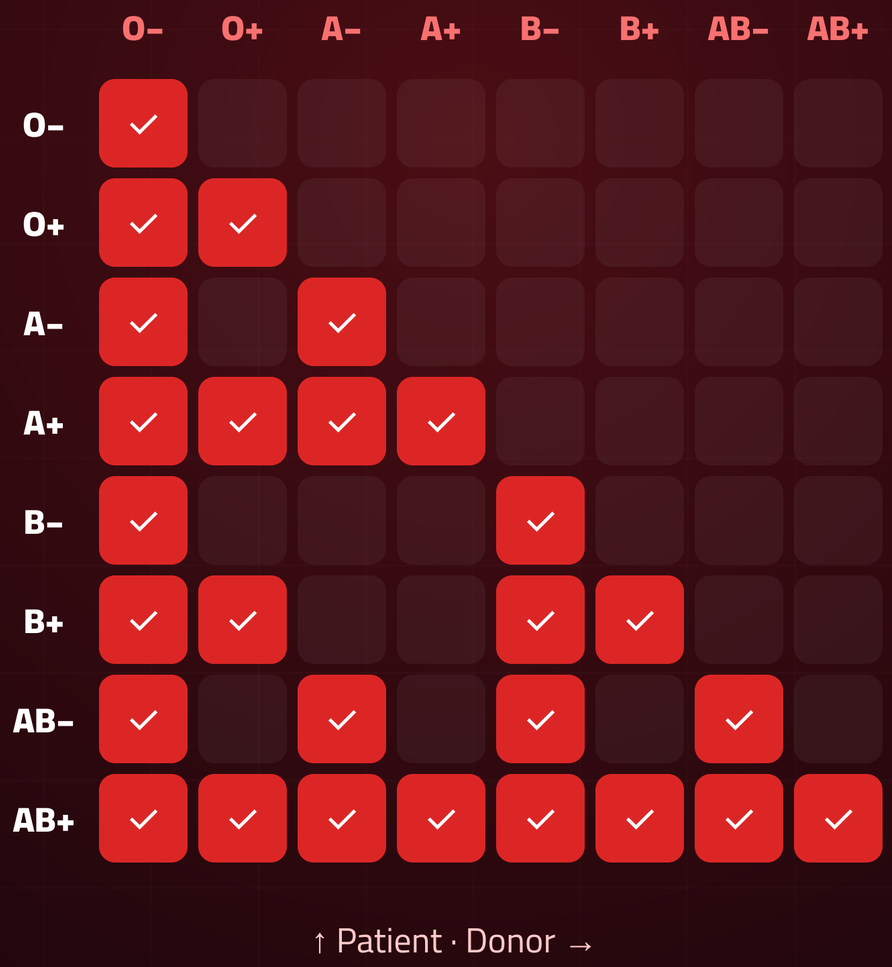


Matching engine: always the right type

O− gives to everyone

AB+ receives from everyone

If a type runs out, it proposes a compatible alternative instantly



The smart donor

✅ **Eligible**

last donation 90+ days ago

📍 **Nearby**

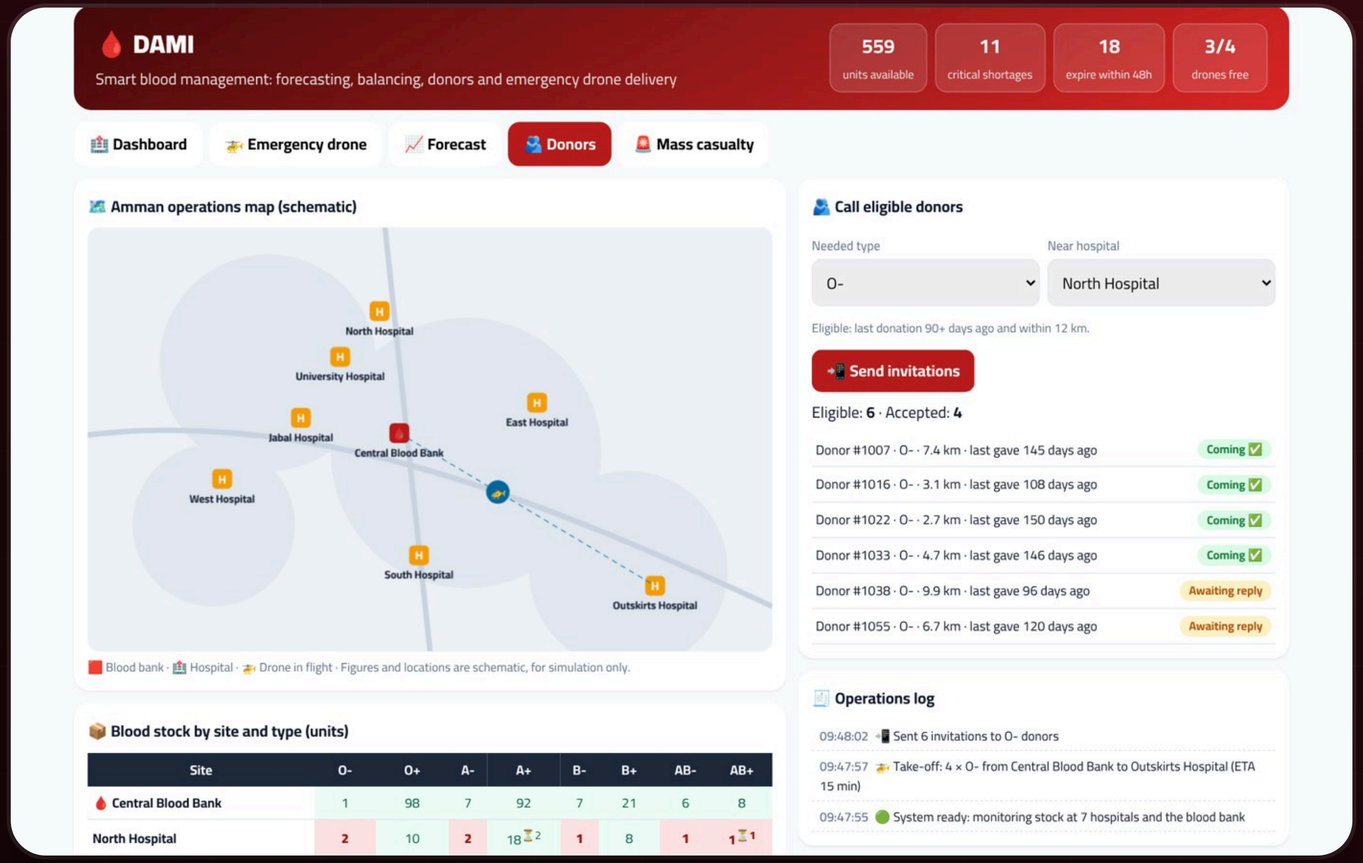
within 12 km of the hospital

🎯 **Targeted**

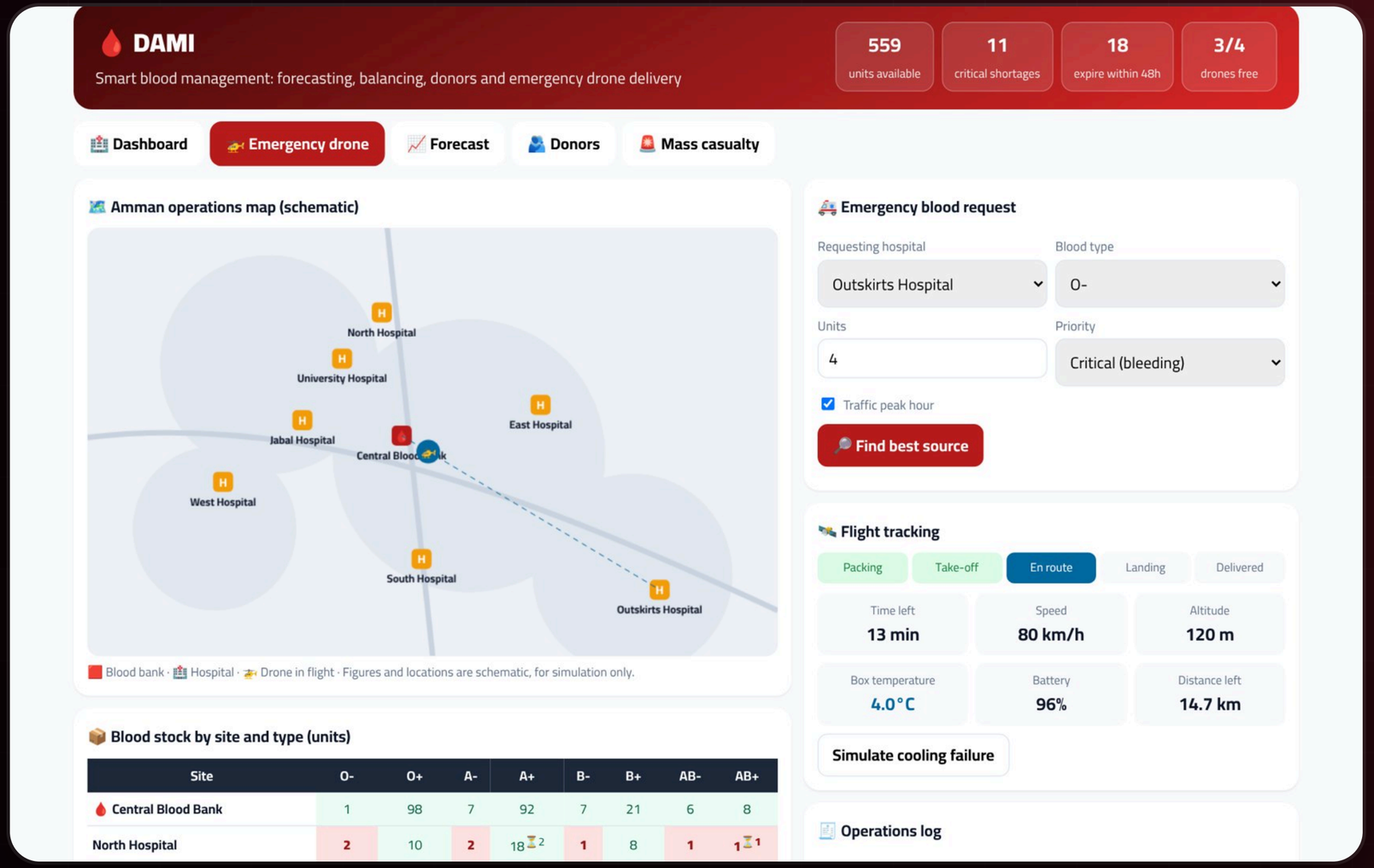
only for the missing type

❤️ **Visible impact**

knows where their unit went



The prototype works today



① **Picks the best source**

② **Compares drone vs car**

③ **Alarms instantly if temperature rises**

Proven worldwide

67%

drop in blood wastage after drone delivery in Rwanda

79 min

faster delivery

🇯🇴 **No similar system in Jordan per our preliminary research**

Source: Nisingizwe et al., The Lancet Global Health, 2022

Market size

TAM

JOD 726K / year

121 hospitals in Jordan

SAM

JOD 426K / year

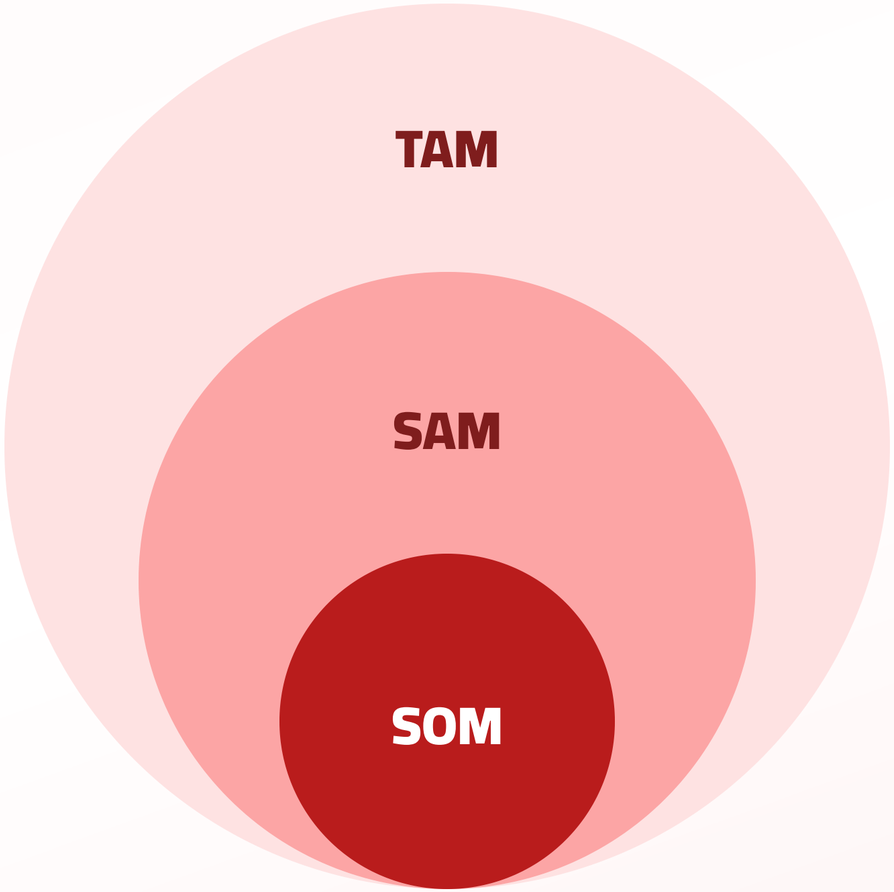
71 private hospitals

SOM

JOD 120K / year

20 hospitals by year 3

+ National dashboard contract · + Emergency flights · + Region: Gulf & Iraq later



Hospital counts: trade.gov — Jordan Country Commercial Guide: Healthcare (121 = 71 private, 33 MoH, 15 RMS, 2 university)

Competition

	Today (manual)	Blood-bank software	Zipline	DAMI
Demand forecasting	✗	~	✗	✓
Stock balancing	✗	✗	✗	✓
Targeted donors	✗	~	✗	✓
Drone delivery	✗	✗	✓	✓
Live cold-chain	✗	~	✓	✓
Built for Jordan	~	~	✗	✓

Based on our preliminary research; ~ = partial

Who pays?



JOD 6,000

hospital subscription / year



JOD 90

per emergency flight



JOD 40,000

national dashboard / year

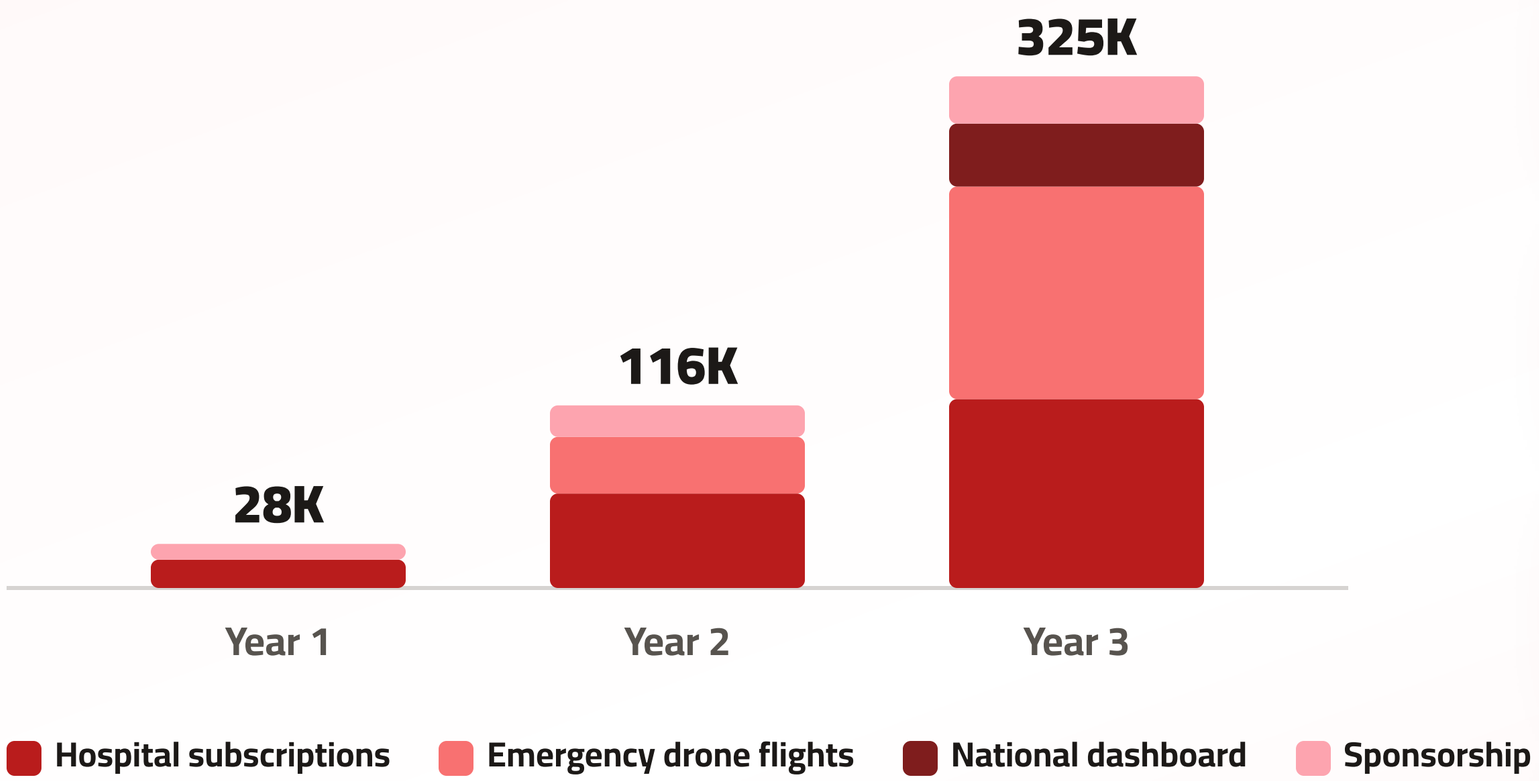


JOD 10,000

donor-app sponsorship / year

72% gross margin in year 3

Revenue: 4 streams, growing



JOD 6,000

per hospital / year

JOD 90

per flight · 55% margin

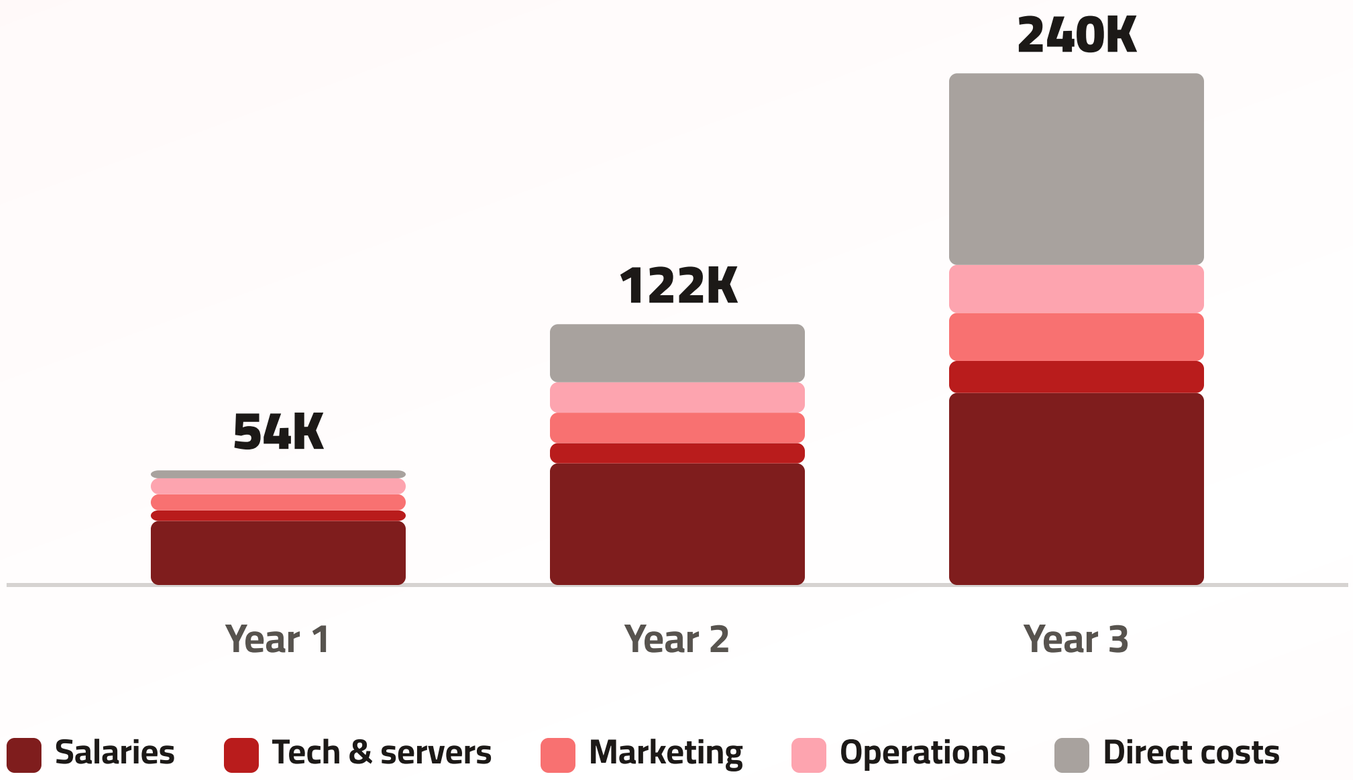
3 → 10 → 20

hospitals

0 → 400 → 1,500

flights

Costs: where the money goes



60%

of OPEX is the team

≈ JOD 40

direct cost per flight

15%

direct cost per subscription

Budget: we need JOD 89,900

30,000 JOD

Drones, cold boxes & ground station

23,000 JOD

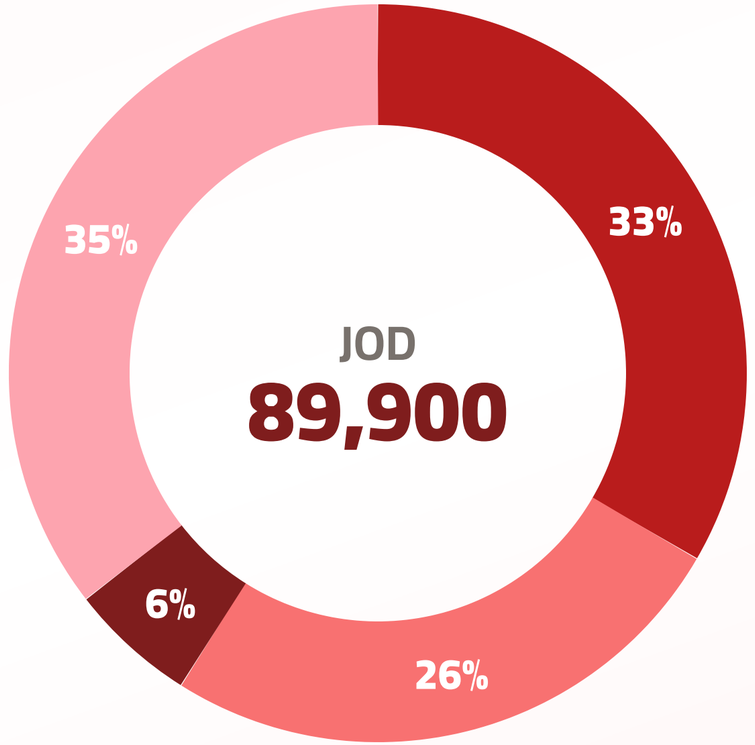
Platform & donor app

5,000 JOD

Security & compliance

31,900 JOD

Operating runway (to month 24)



Covers the initial investment (JOD 58,000) + operating losses until monthly cash flow turns positive in month 25

Financial summary

	Year 1	Year 2	Year 3
Revenue	28.0K	116.0K	325.0K
Direct costs	3.7K	27.2K	89.8K
Gross profit	**24.3K**	**88.8K**	**235.2K**
Operating costs	50.0K	95.0K	150.0K
Operating result	**-25.7K**	**-6.2K**	**85.2K**

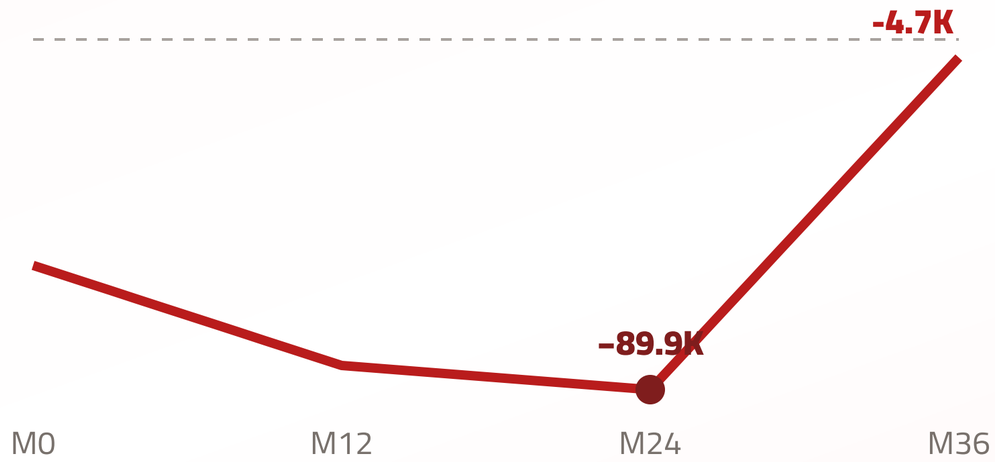
72%
Gross margin Y3

Month 25
Cash-positive from

Month 37
Break-even

Month 27
Optimistic case

Cumulative cash (36 months)



Planning assumptions — to be validated with hospitals and suppliers

The plan

Now

Prototype ✓

6–12 months

3 hospitals

12–24 months

First licensed drone route

24–36 months

National dashboard

Our ask



Seed funding

≈ JOD 90,000



Pilot partnership

blood bank + hospital



Permit support

Civil Aviation Regulatory Commission

Team DAMI · Software Engineering & Artificial Intelligence

Team DAMI

Software Engineering + Artificial Intelligence



Abdulrahman Bilal Aburumman

Software Engineer



Marah Yousef Al-Qaisi

AI Engineer



Nadeen Mohammad Al-Smadi

AI Engineer



Darya Ali Al-Shaqaqha

AI Engineer

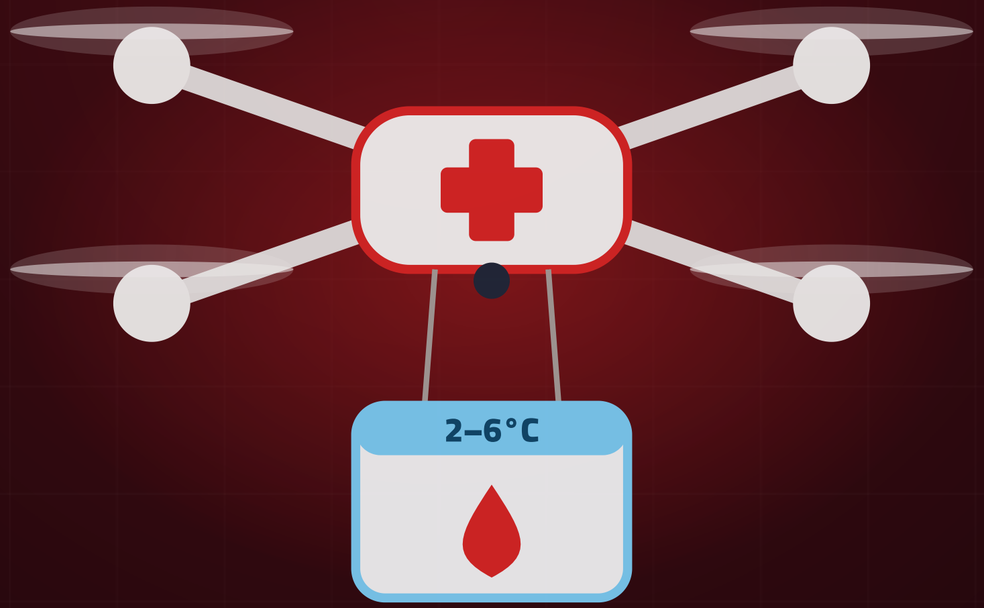


Abdulmonem Fadi Qadan

AI Engineer

Because blood that arrives late...

doesn't arrive.



Team DAMI